TOPIC: THERAPUTIC CANCER VACCINES

BARC INTERNSHIP PPT

INTRODUCTION:

Cancer vaccines are unlike other preventive vaccines like flu shots or HPV vaccine. These common vaccines are given to healthy people as a prevention technique for the future. Therapeutic cancer vaccines are given to patients who already have cancer, with the goal of treating the existing disease.

Rather than directly attacking cancer cells with a drug, like in chemotherapy or surgery, a cancer vaccine trains the patient's own immune system to help recognize and destroy the cancer cells on its own. This is a shift from the broad and general oncology in which a general treatment is given to each patient which isn't personalized and built specifically for the genetic profile of a person's tumor.

HOW ARE THESE THERAPUTIC CANCER VACCINES MADE?[1]

Step 1: The process begins with a biopsy of the patient's tumor. The samples' DNA and RNA are sequences using modern technology which allow scientists to figure out the exact mutations that are unique to the patient's cancer. Computer algorithms then predict which of these mutations, or neoantigens, are most likely to trigger a strong immune response. These programs look at if the mutated protein would actually be seen on the cancer cell's surface or not, helping see if the T-cells could see it or not.

Step 2: After the mutations (neoantigens) are decided, they need to be made into an actual vaccine. This step varies depending on the type of vaccine. Peptide vaccines make the actual mutated proteins in the lab; whereas, mRNA vaccines create a set of genetic instructions which tell the parents' own cells how to make the proteins in the body after injection. The leading mRNA cancer vaccines are – autogene cevueran and mRNA-4157. Both of these use lipid nanoparticles (fatty) capsules for delivery, but differ in design.

1. **Autogene Cevueran**
   This vaccine is given through an IV, directly into the bloodstream, so it can travel to the immune tissue and immune cells. It's made from a natural and unmodified form of mRNA, wrapped in a fatty capsule and carries instructions for around 20 mutations specific to the patient's tumor.
2. **mRNA-4157**
   This vaccine is injected into the muscle and can carry instructions for around 34 patient specific mutations. Unlike autogene cevueran, its mRNA is chemically

[1] https://www.mdpi.com/2076-393X/13/12/1231

altered to help the mRNA last longer in the body and be read more efficiently by the cells.

Step 3: These vaccines train the T-cells of the body to identify these mutations and kill cells with that DNA sequencing.

These vaccines are made from scratch and therefore include a long and expensive process in comparison to the drug that is mass produced for the entire population. Each vaccine made needs to be tested to ensure it meets the safety standards. Therefore, the production time for the vaccine is around 6 to 16 weeks long, which can be dangerous because cancer spreads quickly in the body.

CLINICAL TRIALS

Clinical trials for these therapeutic cancer vaccines have typically been done with last stage cancer patients whose disease was too advanced for surgery, chemotherapy, or immunotherapy. Trials have now moved on to being done after surgery or chemotherapy to prevent the cancer from coming back. Vaccines are usually combined with other treatments rather than used alone.
EG → Melanoma (skin cancer): mRNA-4157 combined with pembrolizumab reduced the risk of recurrence by 44% compared to when pembrolizumab was used on its own.

LIMITATIONS

→These cancer vaccines haven't gained FDA approval yet and are still undergoing clinical trials.
→ Manufacturing of these vaccines is a long, slow, and expensive process. It lasts for around 6 to 16 weeks.
→ Not every neoantigen might now respond like they are supposed to.
→ It's a long and resource-intensive process which means it's difficult to carry it out at a large scale.
→ Certain types of cancer may produce fewer neoantigens which mean these vaccines can be deemed ineffective for those patients.

CONCLUSION

Therapeutic cancer vaccines represent a major shift in the treatment of cancer, from general drugs to vaccines that are personalized for a patient's own tumor. The results from clinical trials are promising but still a lot more ongoing research is being done to confirm the efficacy of these drugs.
As more clinical trials take place, these vaccines may be a very promising way to treat cancer in the future.